

Course Schedule

Schedule is subject to change. All changes are posted here. If a due date here conflicts with Canvas, this page is correct.

Materials link up as they are released; a week with no links is not published yet.

Module 1: Foundations

Why do we care about quantitative representations of climate hazard?

Week	Date	In class	Due
1	Mon 8/24	Welcome to CEVE 543 · Course logistics and expectations	
	Wed 8/26	Return periods and exceedance probabilities · Return periods in Julia	Week 01 Practice Problems (Optional); Week 01 Reading: Motivations (nothing to submit)
	Fri 8/28	Monte Carlo simulation · Monte Carlo in Julia · Lab 1: Monte Carlo convergence	Lab 1 software install
2	Mon 8/31	Intro to the climate system · Climate as a dynamical system	Week 02 Practice Problems (Optional); Week 02 Reading: Climate as a dynamical system
	Wed 9/2	Autocorrelation, partial autocorrelation, and the spectrum	
	Fri 9/4	Lab 2: Climate as a dynamical system	Lab 1 due
3	Mon 9/7	No class (Labor Day)	
	Wed 9/9	Week 3 overview · Bayes, the probability integral transform, and the bootstrap	Practice Problems TBD
	Fri 9/11	Lab 3: Bayes, the probability integral transform, and the bootstrap · Oral format demonstration	Lab 2 due

Module 2: Univariate extremes

How do we characterize the probability distribution of a single hydroclimate hazard at a single location, and what are the associated uncertainties?

Week	Date	In class	Due
4	Mon 9/14	Week 4 overview · The extremal types theorem and the GEV · Block maxima and return levels	Practice Problems TBD; Reading TBD
	Wed 9/16	Fitting a GEV in Julia	
	Fri 9/18	Lab 4: Extreme value theory: GEV, GPD, block maxima and exceedances	Lab 3 due
5	Mon 9/21	Week 5 overview · The GPD and Pickands-Balkema-de Haan · Peaks over threshold in practice	Practice Problems TBD; Reading TBD

Week	Date	In class	Due
	Wed 9/23	Threshold selection and declustering	
	Fri 9/25	Lab 5: Peaks over threshold in practice	Lab 4 due
6	Mon 9/28	Week 6 overview · Nonstationarity: trends, covariates, and change points	Practice Problems TBD; Reading TBD
	Wed 9/30	Time-varying GEV and trend tests	
	Fri 10/2	Lab 6: Nonstationarity: trends, covariates, and change points	Lab 5 due
7	Mon 10/5	Week 7 overview · L-moments and the index-flood model · Regional frequency analysis and partial pooling	Practice Problems TBD; Reading TBD; Topic menu published
	Wed 10/7	Hierarchical models in Turing.jl	
	Fri 10/9	Lab 7: Regional frequency analysis and partial pooling	Lab 6 due

Module 3: Spatiotemporal extremes

How do we sample hydroclimate hazard(s) across space and/or time to characterize complex extremes?

Week	Date	In class	Due
8	Mon 10/12	No class (Midterm recess)	
	Wed 10/14	Week 8 overview · Model selection	Practice Problems TBD; Reading TBD
	Fri 10/16	Test 1	Lab 7 due
9	Mon 10/19	Week 9 overview · Sklar's theorem and tail dependence · Joint distributions and the Gaussian copula	Practice Problems TBD; Reading TBD
	Wed 10/21	Simulating dependent extremes	
	Fri 10/23	Lab 8: Joint distributions and the Gaussian copula	
10	Mon 10/26	Week 10 overview · Storm transposition	Practice Problems TBD; Reading TBD; Selection due
	Wed 10/28	The transposition integral in practice	
	Fri 10/30	Lab 9: Storm transposition	Lab 8 due
11	Mon 11/2	Week 11 overview · Climate models, downscaling, and bias correction	Practice Problems TBD; Reading TBD
	Wed 11/4	Quantile mapping and delta methods	
	Fri 11/6	Lab 10: Climate models, downscaling, and bias correction	Lab 9 due
12	Mon 11/9	Week 12 overview · PCA and EOF analysis · Weather typing	Practice Problems TBD; Reading TBD
	Wed 11/11	PCA and k-means on reanalysis fields	
	Fri 11/13	Lab 11: Weather typing	Lab 10 due
13	Mon 11/16	Week 13 overview · Markov chains and hidden Markov models · Climate-informed generators	Practice Problems TBD; Reading TBD
	Wed 11/18	Building a synthetic generator	Project checkpoint
	Fri 11/20	Lab 12: Climate-informed generators	Lab 11 due

Week	Date	In class	Due
14	Mon 11/23	Peer review of project methods	
	Wed 11/25	No class (Thanksgiving)	
	Fri 11/27	No class (Thanksgiving)	

Synthesis

Week	Date	In class	Due
15	Mon 11/30	Test 2	Lab 12 due
	Wed 12/2	Discussion	
	Fri 12/4	Synthesis and wrap-up	Written version due

Bibliography